DEPARMENT OF INFORMATION TECHNOLOGY
FACULTY OF ENGINEERING AND TECHNOLOGY

IT25-425 P-II
COMPUTER NETWORKS
MORNING/EVENING

LAB#6

TASK: CONFIGURATION OF A DEDICATED DHCP SERVER

Name: _____ Roll No:_____

Score: _____ Signature of the Lab Tutor _____ Date: _____

Objectives:

To configure a dedicated DHCP server in Cisco Packet Tracer, assign a static IP to the server, enable the DHCP service with your network pool settings, and connect end devices via a switch.

Apparatus (Software): Packet Tracer 9.01

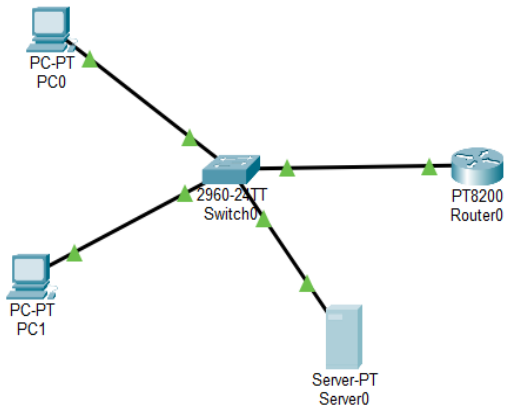
Procedure:

Step 1: Set Up the Network Topology

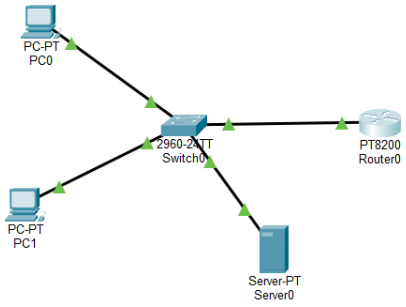
i. Add a Switch (2960), a Router (PT8200 or similar), and a few PCs to your workspace.

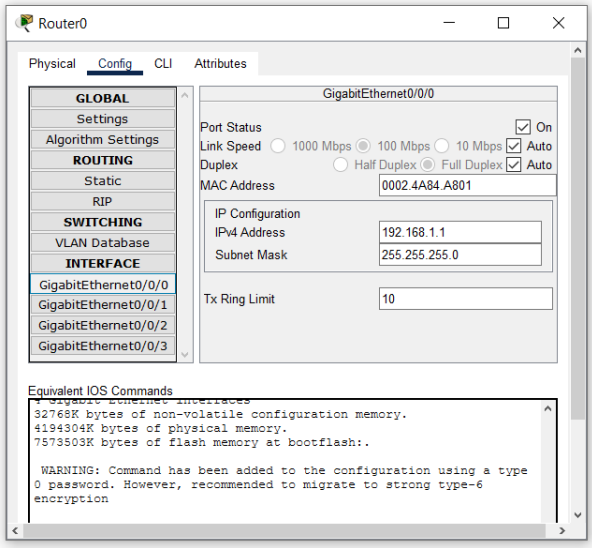
ii. Connect the Server, Router, and PCs to the Switch using Copper Straight-Through cables.

iii. Connect the Router to the Switch interface that serves as your gateway.



iv. Click the Router go to the configuration select the GigabitEthernet0 0/0 turn the port status ON.
v. Assign the IP address: `192.168.1.1`, Subnet Mask: `255.255.255.0` to the Router Interface

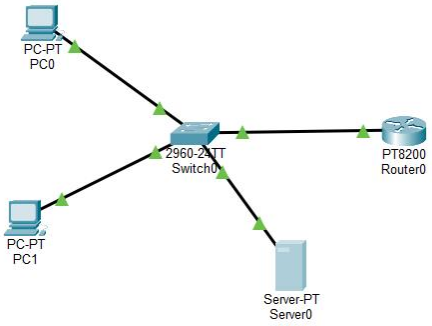


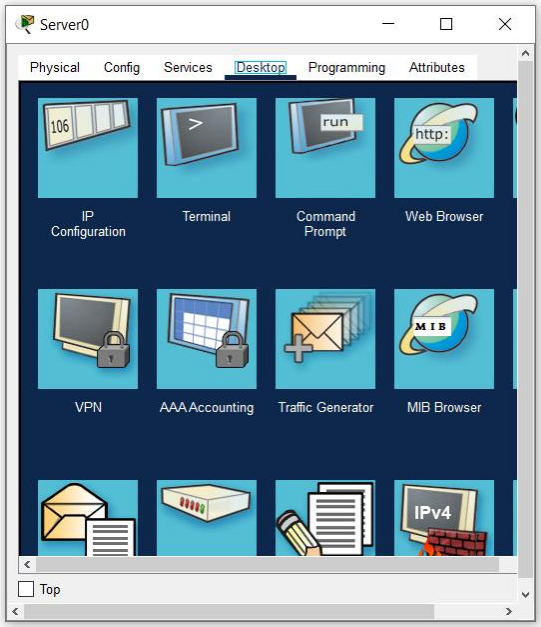


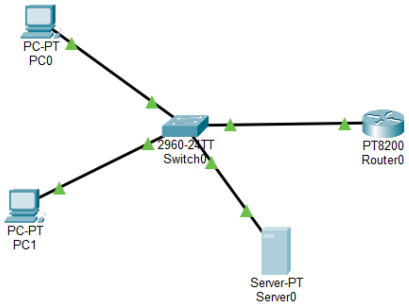
Step 2: Configure the Server IP Address

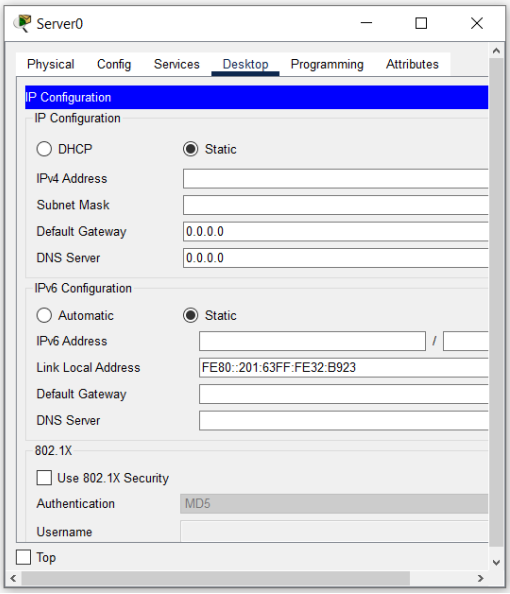
Now the main part star from Server

a. Click on the Server and open the Desktop tab.
b. Click IP Configuration.

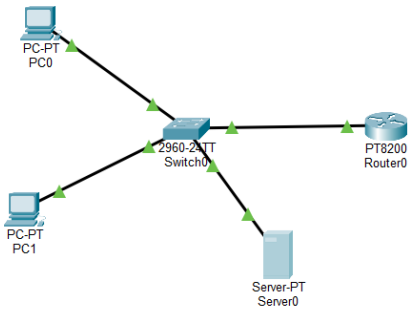


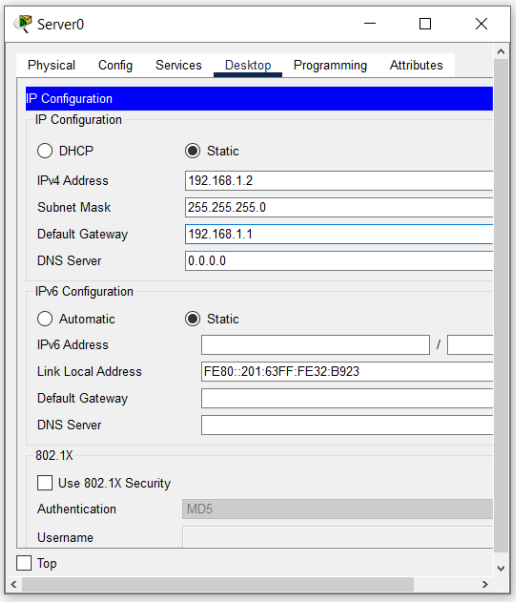






c. Assign a static IP address to the server:
   - IPv4 Address: `192.168.1.2`
   - Subnet Mask: `255.255.255.0`
   - Default Gateway: `192.168.1.1` (matches your router interface IP)

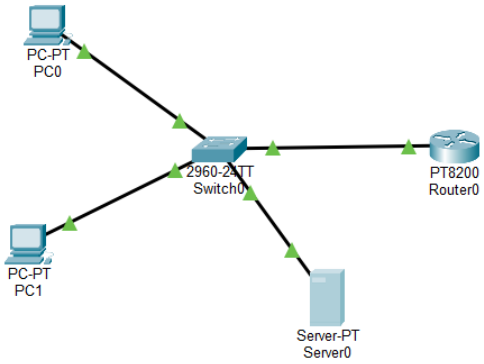


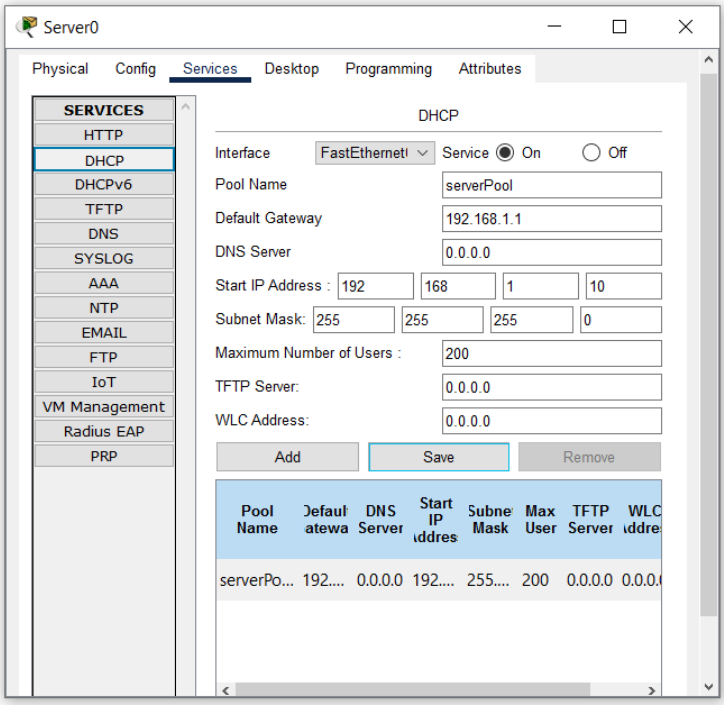


d. Close the IP configuration window.

Step 3: Enable and Configure DHCP Service

- Stay in the Server window and click the Services tab.
- Select DHCP from the left-hand menu.
- Turn the service ON by clicking the radio button.
- Fill in the pool parameters:
- Pool Name: `ServerPool` (or `LAN`)
- Default Gateway: `192.168.1.1`
- Start IP Address: `192.168.1.10`
- Subnet Mask: `255.255.255.0`
- Maximum Number of Users `200`

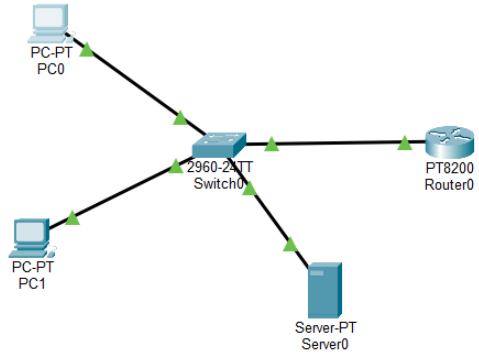


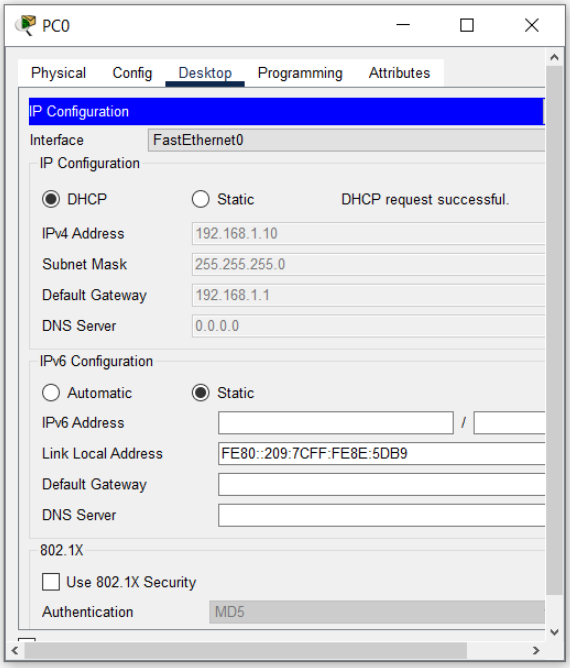


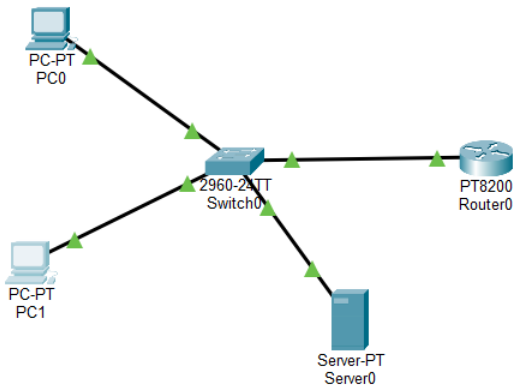
- Click Add or Save to update the pool list.

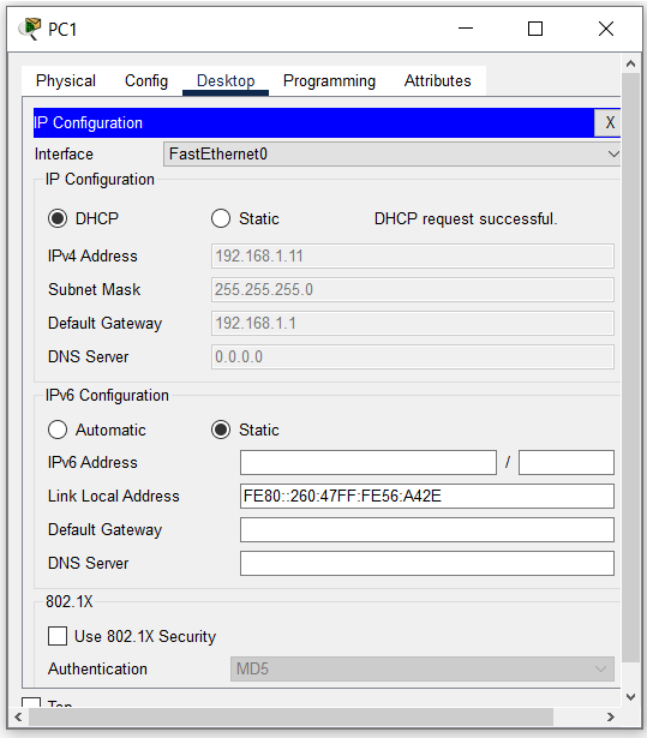
Step 4: Configure the PC's DHCP Option

- Now go to the PCs, click the DHCP option and see if the IP addresses are automatically assigned or not.







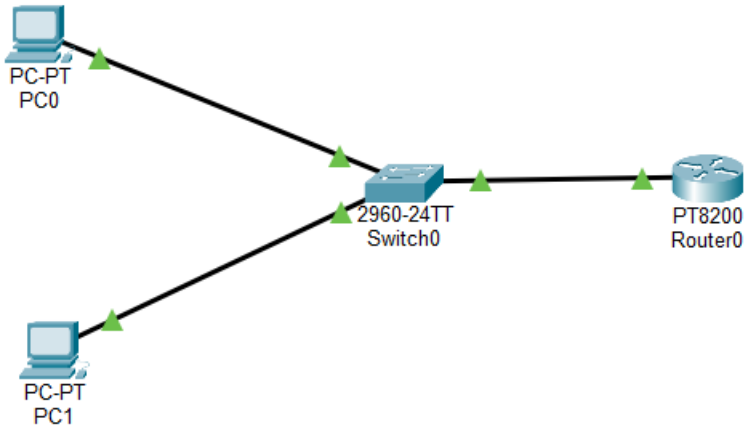


Configuration of Router as a DHCP Server

Step 1: Set Up the Network Topology

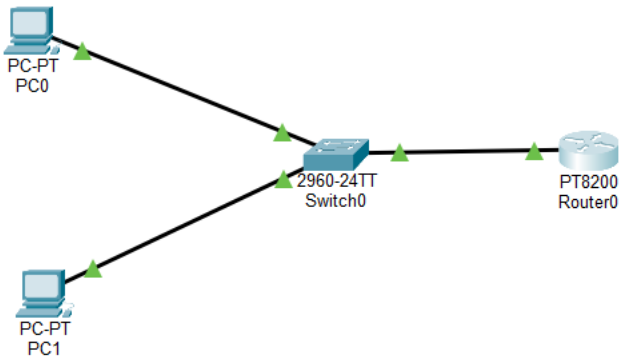
i. Add a Switch (2960), a Router (PT8200 or similar), and a few PCs to your workspace.

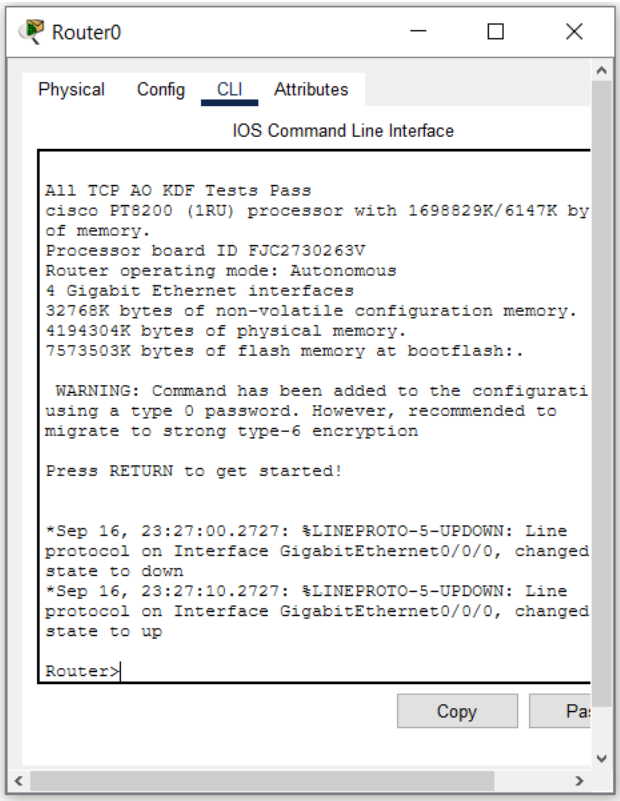
ii. Connect the Router, and PCs to the Switch using Copper Straight-Through cables.



Step 2: Set Up the Router Interface and IP address

Click on the router0, go to the mainline interface and click CLI tab.
Continue with Configuration dialog? (yes/no) No
Press No because we will manually configure the router with commands.





To enter into the privileged exec mode type:
Router> enable or
Router>en
You will enter into the privileged exec mode when you see router followed by # sign
Router#
To enter in to the global configuration mode type:
Router #config t
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#interface GigabitEthernet0/0/0
Router(config-if)#ip add 192.168.1.1 255.255.255.0
Now, enable the interface type or turn the port up: write the following command
Router(config-if)#no shutdown
Router(config-if)#exit

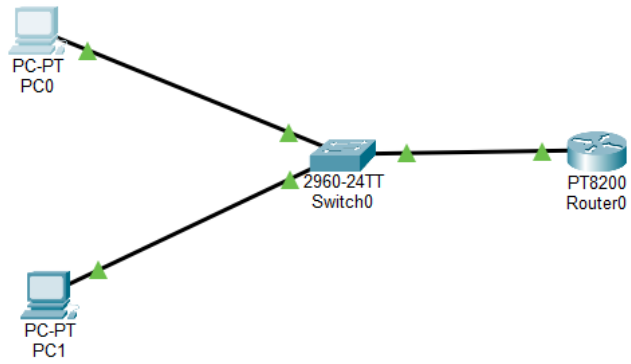
Step 3: Configuration of DHCP Server on a Router

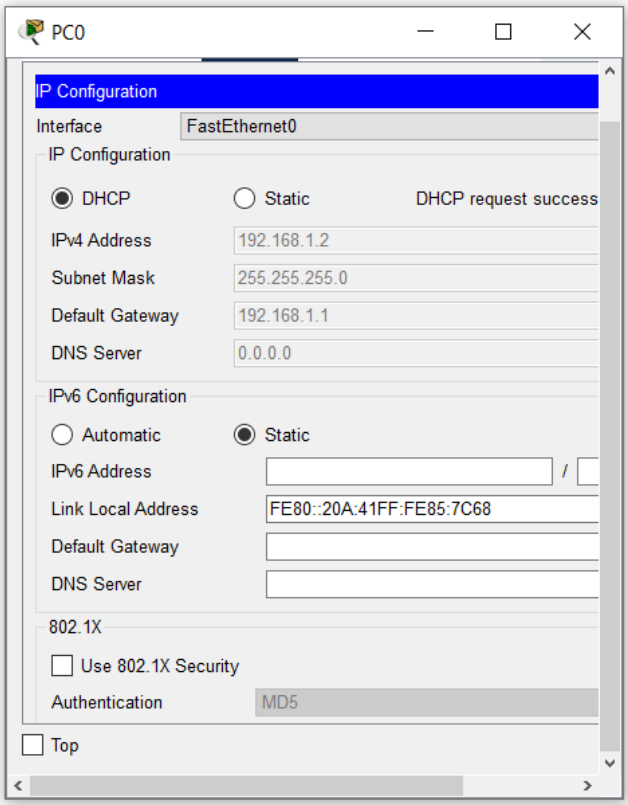
Here you will assign a DHCP Pool IP Addresses

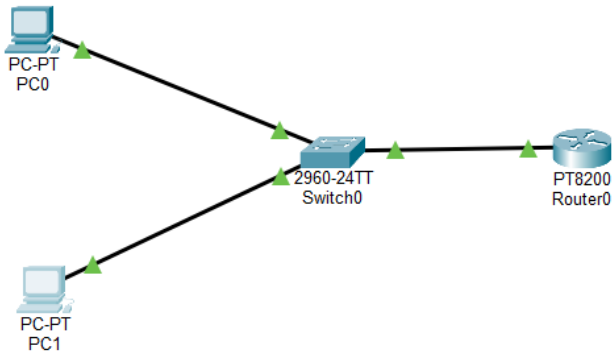
Router(config)#ip dhcp pool mypool
Router(dhcp-config)#network 192.168.1.0 255.255.255.0
Router(dhcp-config)#default-router 192.168.1.1

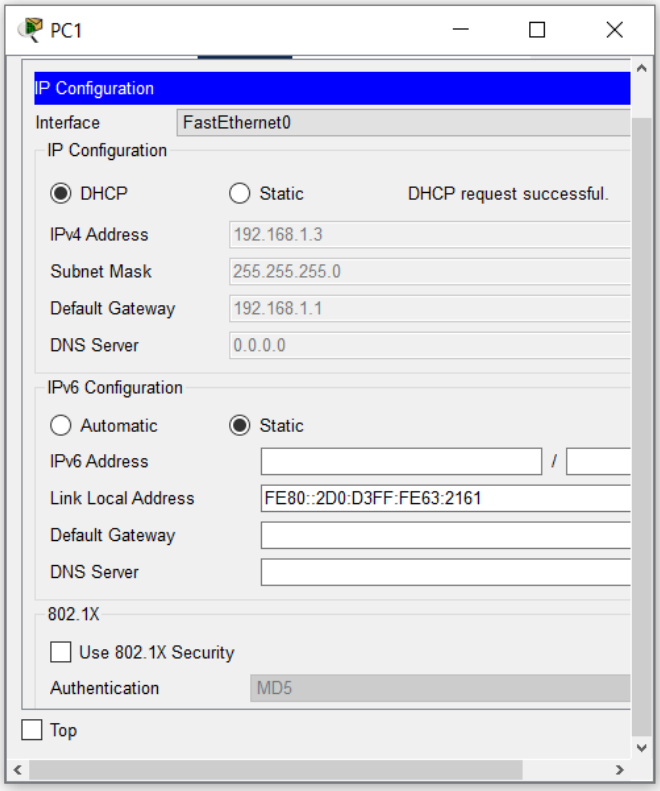
Step 4: Configure the PC's DHCP Option

- Now go to the PCs, click the DHCP option and see if the IP addresses are automatically assigned or not.









Review Questions

Question 1: What is DHCP and How Does It Work?

Answer:

Question 2 Purpose of DHCP Pool is:

Answer:

Question 3: How do you configure a Cisco router as a DHCP server?

Answer:

Note: Submit the handout before next lab